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Simulation of the Bursting Activity of Neuron R15 in *Aplysia*: Role of Ionic Currents, Calcium Balance, and Modulatory Transmitters

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SUMMARY AND CONCLUSIONS

1. An equivalent circuit model of the R15 bursting neuron in *Aplysia* has been combined with a fluid compartment model, resulting in a model that incorporates descriptions of most of the membrane ion channels that are known to exist in the somata of R15, as well as providing a Ca^{2+} balance on the cell.

2. A voltage-activated, calcium-inactivated Ca^{2+} current (denoted the slow inward current I_{SI}) was sufficient to produce bursting activity without invoking any other calcium-dependent currents (such as a nonspecific cation current, I_{NS}, or a calcium-activated K^+ current, $I_{K,Ca}$). Furthermore, many characteristics of a typical R15 burst could be simulated, such as a parabolic variation in interspike interval, the depolarizing afterpotential (DAP), and the progressive decrease in the undershoots of spikes during a burst.

3. The dynamic activity of R15 was analyzed by separately characterizing two different temporal domains: the fast dynamics associated with action potentials and the slow dynamics associated with low-amplitude oscillations lasting tens of seconds ("slow waves"). The slow dynamics were isolated by setting the Na^+ conductance ($\bar{g}_{Na}$) to zero and then studied by the use of a system of equations reduced to two variables: intracellular concentration of Ca^{2+} and membrane potential. The fixed point of the system was located at the intersection of the nullclines for these two variables. A stability analysis of the fixed point was then used to determine whether a given set of parameters would produce slow-wave activity.

4. If the reduced model predicted slow-wave oscillations for a given set of parameters with $\bar{g}_{Na}$ set to zero, then bursting activity was observed for the same set of parameters in the full model with $\bar{g}_{Na}$ reset to its control value. However, for certain sets of parameters with $\bar{g}_{Na}$ at its usual value, the full model exhibited bursting activity because of a slow oscillation produced by the activation of I_{NS} by action potentials. This oscillation resulted from an interaction between the fast and slow dynamics that the reduced model alone could not predict and was not observed when $\bar{g}_{Na}$ was subsequently set to zero. If $\bar{g}_{NS}$ was also set to zero, this discrepancy disappeared.

5. The model predicted a number of experimentally observed transitions in the dynamic activity, including the following: *1*) the transition from a bursting mode to slow-wave activity induced by the Na^+ channel blocker tetrodotoxin (TTX), *2*) the transition from a bursting to a beating mode induced by the Na^+-K^+ pump blocker oubain, and *3*) the entire range of transitions from a hyperpolarized mode through bursting modes to a beating mode produced by external current injection.

6. The model also simulated the effects of the modulatory agents serotonin (5-HT), dopamine, FMRFamide (Phe-Met-Arg-Phe-amide), and egg-laying hormone (ELH), and the second messenger guanosine 3′,5′-cyclic monophosphate (cGMP). These agents modulate either I_{SI} or the anomalous rectifier current (I_R) or both. A critical distinction between the modulation of these two currents was that only I_{SI} affected both the calcium nullcline and the potential nullcline. This distinction provided an alternative mechanism for the paradoxical effects of reducing I_{SI}. Specifically, depending on the new location of the fixed point, a reduction in I_{SI} can either hyperpolarize the cell or induce slow beating.

INTRODUCTION

The R15 cell in the abdominal ganglion of *Aplysia* has been used extensively to gain insights into the biophysical mechanisms underlying endogenous bursting activity in nerve cells. Consequently, a good deal is known about both the specific membrane currents that underlie individual action potentials and the currents that contribute to the slower oscillation, which modulates bursting activity (e.g., Adams 1985; Adams and Benson 1985; Adams and Gage 1979a,b; Adams and Levitan 1985; Gorman et al. 1982). A number of Hodgkin-Huxley equivalent circuit models of R15 have been developed that simulate aspects of the activity of R15 (Adams and Benson 1985; Chay 1983, 1990; Chay and Cook 1988; Hindmarsh and Rose 1984; Plant 1981; Plant and Kim 1976; Rinzel and Lee 1987). These models, however, do not simulate the actual magnitudes and time courses of the individual currents, and only the more recent models (Chay and Cook 1988; Rinzel and Lee 1987) include a description of the calcium-dependent inactivation of a Ca^{2+} current, which is now believed to be essential for bursting activity (Adams and Levitan 1985). In addition, none of the previous models contain a Ca^{2+} balance.

Our objective was to simulate the wide range of electrochemical activity of R15 in a manner consistent with available data on the electrophysiological properties and bulk Ca^{2+} concentrations. The following description enumerates the aspects of the activity of R15 that our model is designed to simulate. A prominent feature of R15 is that it exhibits endogenous bursting activity under conditions for which the interburst potential level is between −70 and −30 mV (Mathieu and Roberge 1971). When the potential is more negative than −70 mV, the cell is silent (Mathieu and Roberge 1971), whereas as the interburst potential approaches −30 mV, the cell undergoes a transition to the production of action potentials continuously at a regular rate (this mode is sometimes referred to as "beating") (Junge and Stephens 1973). In the bursting mode, action potentials produced during a burst exhibit three types of nonuniformity. First, the spike height initially increases and then remains constant. Second, the undershoot de-

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creases. Third, the spike frequency increases then decreases, which results in a "parabolic" bursting pattern (Faber and Klee 1972; Strumwasser 1967). Each burst is followed by a depolarizing afterpotential (DAP) that does not reach the spike threshold (Adams 1985). During bursting activity, there is a slow cyclic variation in the concentration of intracellular free Ca^{2+} (Gorman et al. 1981; Gorman and Thomas 1978). A cyclic variation in membrane potential on a similar time scale can be observed if the spike discharges during bursting are abolished by the application of the Na^+ channel blocker tetrodotoxin (TTX) (Mathieu and Roberge 1971; Strumwasser 1968). The model developed in this study simulates the features described above and explains each one in terms of the nonlinear dynamics of the individual currents acting in concert with the mechanisms for the regulation of levels of intracellular Ca^{2+}.

MODEL FORMULATION

Membrane model

The soma of R15 is modeled as a sphere 250 μm diam, which has a surface area of 0.02 cm^2 (Adams and Gage 1979b; Gorman and Hermann 1982). This surface area includes a factor of 10 to account for membrane invagination (Adams and Gage 1979b; Gorman and Mirolli 1972; Mirolli and Talbott 1972). The membrane capacitance was set to 17.5 nF for the cell on the basis of measured data (Adams and Gage 1979b; Kramer and Zucker 1985), which corresponds to a specific capacitance of 0.875 $\mu F/cm^2$. Figure 1 shows a lumped electrical equivalent circuit for the somatic membrane. The model includes the ionic currents responsible for the fast dynamics of the action potentials as well as pump, exchanger, and background currents that contribute to the slow oscillations that appear as subthreshold fluctuations in the membrane potential. Under space-clamp conditions, the differential equation describing the time-dependent changes in membrane potential (V) is

$$\dot{V} = -(I_{Na} + I_{Ca} + I_{SI} + I_{NS} + I_K + I_R + I_L + I_{NaCa} + I_{NaK} + I_{CaP} - I_{STIM})/C_m \quad (1)$$

The fast Na^+ current (I_{Na}), the fast Ca^{2+} current (I_{Ca}), and the delayed rectifier current (I_K) are modeled as voltage-gated ionic currents. The slow inward Ca^{2+} current (I_{SI}) and the nonspecific cation current (I_{NS}) are also modeled as voltage-gated currents, but the description for these currents includes a Ca^{2+}-dependent term as well. The remaining currents are modeled as background currents, hence they are instantaneous. The anomalous rectifier current (I_R) and the leakage current (I_L) are solely dependent on membrane voltage, the Na^+-Ca^{2+} exchanger current (I_{NaCa}) depends on both Ca^{2+} concentration and voltage, the Na^+-K^+ pump current (I_{NaK}) is modeled as a constant, and the Ca^{2+} pump current (I_{CaP}) has only a Ca^{2+} concentration dependence. I_{STIM} is the applied external stimulus current. A full set of model equations is given in the APPENDIX, where the Hodgkin-Huxley type gating variables (m, h, n, l, d, f, s, and b) are described by solutions of first-order differential equations of the general form

$$\dot{z} = (z_\infty - z)/\tau_z \quad (2)$$

where z_∞ and τ_z are the steady-state value and the time constant, respectively, associated with the specific gating variable.

Full descriptions of the membrane currents are provided below. In general, we have attempted to utilize published experimental data on R15. In some cases, however, there was insufficient data in the literature, and our approach in such cases has been to adopt an appropriate mathematical form from the literature, and then adjust the parameters of the equations in a manner consistent with the available information. The mathematical equations representing these currents are given in glossary form in the APPENDIX. The transient-outward K^+ current (I_A), which is largely inactivated in the normal voltage range of the cell (Benson and Levitan 1983), was not included in the model; neither was the Ca^{2+}-activated K^+ current ($I_{K,Ca}$). $I_{K,Ca}$, together with some justification for its omission, is treated in the DISCUSSION.

Inward currents

I_{Na}. The properties of the fast Na^+ channel (I_{Na}) have been analyzed by Adams and Gage (1979a,b). Analytic expressions for the steady-state values and time constants associated with the gating variables m and h were obtained by using a nonlinear least-squares

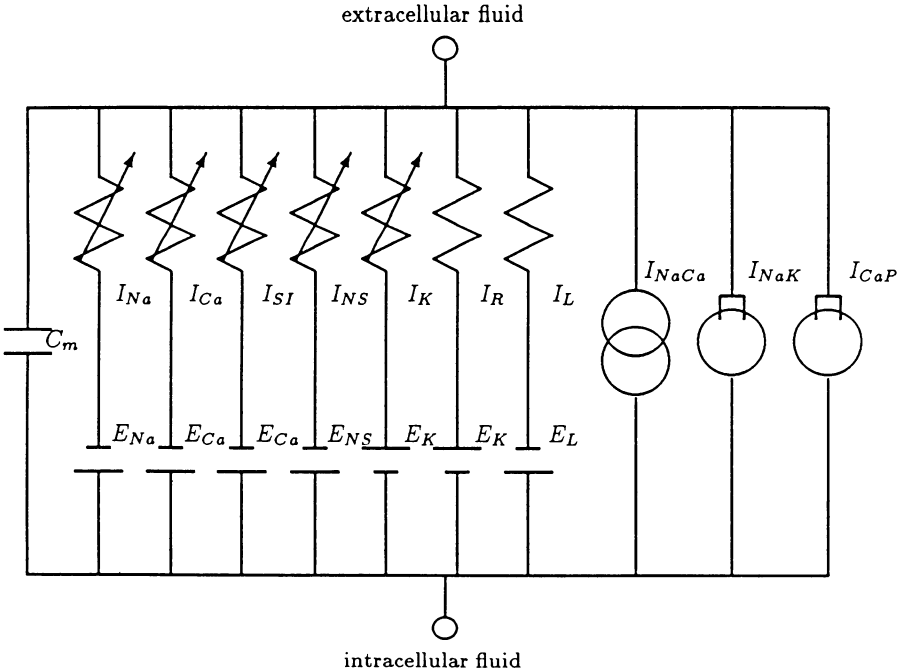


FIG. 1. Equivalent circuit diagram for the model.

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parameter estimation algorithm (Marquardt 1963) to fit their data (Adams and Gage 1979a). Rate constants were scaled from 13.5 to 22°C by the use of a Q_{10} of 2 (Adams and Gage 1979b), and the maximum Na^+ conductance was estimated to be 30 μS for the cell, or 1.5 mS/cm^2 (Adams and Gage 1979a; Gorman and Hermann 1982). To produce sustained bursts that closely simulated typical experimental data from R15, we found it necessary to shift the half-inactivation of h_∞ (see Fig. 2) from −32 mV as reported by Adams and Gage (1979a) to 22 mV (see also Adams and Benson 1985).

I_{Ca}. I_{Ca} (Adams 1985; Adams and Gage 1979a) contributes to the production of spikes and should be distinguished from a second Ca^{2+} current called the I_{SI} (see below), which has a smaller magnitude, is activated at more negative potentials, and contributes to the production of the slow potential oscillations. The specific description for I_{Ca} was obtained by a Marquart fit to data of Adams and Gage (1979a) for the voltage-dependent activation and inactivation gating variables. The activation variable was fit to a d^2 form (Adams and Gage 1979a), the rate constants were scaled from 13.5 to 22°C by the use of a Q_{10} of 2 (Adams and Gage 1979b), and the conductance of the channel was estimated as 10 μS for the cell, or 0.5 mS/cm^2 (Adams and Gage 1979a; Gorman and Hermann 1982). The reversal potential for the Ca^{2+} currents (E_{Ca}) was fixed at the experimentally observed value of 65 mV (Adams and Gage 1979a). We used the analytic description of the steady-state value of the inactivation gating variable f as described by Adams and Gage (1979a). This expression may only be an approximation, however, because more recent studies (Eckert and Chad 1984) indicated that this channel may exhibit Ca^{2+}-dependent inactivation, rather than voltage-dependent inactivation. Given the rapid inactivation of this channel relative to the time scales of burst modulation, either type of inactivation should give roughly the same results. In both cases, inactivation occurs primarily at potentials above the threshold for I_{Ca}, when the Ca^{2+} concentration in the immediate vicinity of the channel is elevated because of the influx of Ca^{2+}, and recovers from inactivation below the threshold.

I_{SI}. I_{SI} is a key current responsible for the bursting activity in the model, as well as for the region of negative slope resistance [in the current-voltage (I-V) relationship], which is frequently associated with bursting activity. The critical aspect of this current is that it is both voltage activated and Ca^{2+} inactivated. There is some controversy over which ion(s) carries this current (Adams and Benson 1985; Adams and Levitan 1985; Johnston 1976; Smith et al. 1975). Our description of I_{SI} is based on the assumption that I_{SI} is carried only by Ca^{2+} ions (Adams and Levitan 1985; Gorman et al. 1982).

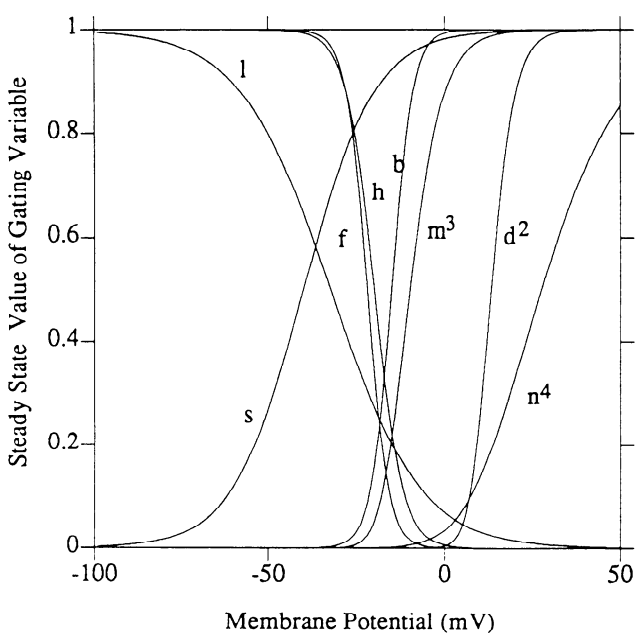


FIG. 2. Gating parameters of the model. Steady-state values of the gating variables are plotted vs. potential. The variables m^3 and h are associated with I_{Na}; n^4 and l with I_K; d^2 and f with I_{Ca}; b with I_{NS}; and s with I_{SI}.

A slowly activating, persistent inward current has been observed during long-lasting voltage-clamp pulses by Gola (1974) in R15 and by others in *Helix* neurons (Eckert and Lux 1975). This current also exhibits even slower, incomplete inactivation (Ehile and Gola 1979). Because this current is small in amplitude and difficult to study in isolation, voltage-clamp data are incomplete [see DISCUSSION for the relationship of this current to the currents in the literature referred to as I_{NSR} (Adams and Benson 1985), I_B (Adams et al. 1980), and I_D and I_H (Adams 1985)]. In lieu of using published voltage-clamp data in a parameter estimation algorithm, we adjusted the applicable parameters by hand to produce a description of I_{SI} consistent with the following observations. First, I_{SI} has a low threshold (−60 mV or less) and activates steeply between −50 and −20 mV (Adams and Benson 1985; Gorman and Hermann 1982; Gorman et al. 1982; Kramer and Zucker 1985). The steady-state activation (s_∞) curve was generated accordingly (see Fig. 2). Second, Gola (1974) reported a time constant of activation (τ_s) on the order of 1 s in the potential range from −50 to −35 mV. We found that a description of the voltage dependence of τ_s as a bell-shaped function with a peak of 2 s at about −25 mV produced the most accurate simulations of bursting activity in R15 [but see Adams et al. (1980) in which a decrease from 2 s at −60 mV to 200 ms at 0 mV was described as typical for molluscan neurons]. Third, the slow inactivation (on the order of tens of seconds) of this current was attributed to Ca^{2+}-dependent inactivation (Adams 1985; Adams and Levitan 1985; Kramer and Zucker 1985). We chose a half-inactivation level (0.025 μM) that is well below the resting level (at −50 mV) of 0.2 μM (Gorman et al. 1981; Gorman and Thomas 1978) to produce a nearly linear dependence of the inactivation on the change in intracellular levels of Ca^{2+}. Finally, the conductance of this channel was adjusted to produce a current on the order of 5 nA at physiological levels of Ca^{2+} (Adams and Levitan 1985; Kramer and Zucker 1985).

I_{NS}. We included a nonspecific cation current to better simulate the depolarizing aftercurrent (I_D) (Adams 1985; Adams and Levitan 1985; Kramer and Zucker 1985; Thompson and Smith 1976; Zucker 1988) seen in molluscan bursting neurons, although the exact nature of the depolarizing aftercurrent in R15 is controversial (see DISCUSSION). The essential features of I_{NS} are that it is an inward current (at potentials below the threshold for spike initiation) that is activated by action potentials and summates during a burst, thus boosting the spiking frequency. In bursting neurons in *Aplysia*, I_{NS} is carried by Na^+, Ca^{2+}, and K^+ ions and has a reversal potential of about −22 mV (Kramer and Zucker 1985; Zucker 1988). In contrast to I_{SI} (which is Ca^{2+} inactivated), I_{NS} is Ca^{2+} activated. We selected a half-activation concentration level in the midrange of $[Ca]_i$ levels generated by model simulations of bursting. In addition, we included a sigmoidal activation function (b_∞, see Fig. 2), which yielded full activation over a 10-mV range from −20 to −10 mV (Adams and Levitan 1985; but see also Lewis 1984; Zucker 1988), and a voltage-dependent time constant, which allowed the activation to summate during a burst.

To include this current in the calcium balance, we estimated the fraction of I_{NS} carried by Ca^{2+} ions on the basis of the following two assumptions. First, the nonspecific cation channel behaves as a sum of Ca^{2+}, Na^+, and K^+ channels with identical concentration dependence and gating characteristics. This assumption can be stated as follows

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$$\bar{g}_{NS}(V - E_{NS}) = \bar{g}_{NS,Ca}(V - E_{Ca}) + \bar{g}_{NS,Na}(V - E_{Na}) + \bar{g}_{NS,K}(V - E_K) \quad (3)$$

Thus the sum of the component conductances must equal $\bar{g}_{NS}$, and the sum of the products of the component conductances with their respective reversal potentials must total $\bar{g}_{NS}E_{NS}$. A second assumption, that $\bar{g}_{NS,Ca}$ equals $\bar{g}_{NS,Na}$, completes a set of three equations in three unknowns that can be solved for the three component conductances. The appropriate multiplier for I_{NS} in the calcium balance (see APPENDIX, *Internal calcium concentration*) is given by $[\bar{g}_{NS,Ca}(V - E_{Ca})]/[\bar{g}_{NS}(V - E_{NS})]$, where the calculated value for $\bar{g}_{NS,Ca}$ is 0.197 μS.

I_L. Several currents that are normally included in the measurement of the instantaneous leakage current (such as I_R, I_{NaCa}, I_{CaP}, and particularly I_{NaK}) have been modeled separately, so we found it necessary to set the reversal potential E_L to 0 mV rather than a more typical value of −45 mV (Gorman and Hermann 1982) to preserve the character of the steady-state *I-V* curve (see RESULTS).

Outward currents

I_K. The description for I_K was obtained by using a Marquart parameter estimation algorithm to fit voltage-clamp data of Hermann and Gorman (1981a,b) to a Hodgkin-Huxley format utilizing an activation variable n raised to the fourth power and an inactivation variable l [results were scaled from 16 to 22°C by the use of a Q_{10} of 2 (Adams and Gage 1979b)]. The mathematical form for the time constant of inactivation was taken from the L14 cell of *Aplysia* because comparable data for R15 were not available (Byrne 1980a,b). The maximum delayed rectifier conductance $\bar{g}_K$ was estimated at 70 μS for the cell, or 3.5 mS/cm^2 (Adams and Gage 1979a; Gorman and Hermann 1982).

I_R. I_R is the dominant current at very hyperpolarized potentials (below −70 mV) (Benson and Adams 1987). The mathematical form utilized to describe this current was adapted from an analysis on starfish egg cells (Hagiwara and Takahashi 1974). Some of the parameters of the mathematical description were adjusted to fit the experimentally determined reversal potential and approximate magnitude of I_R (Adams and Benson 1985; Benson and Adams 1987; Benson and Levitan 1983) at an extracellular concentration of 10 mM K^+ (see APPENDIX, *Outward currents*).

Pumps and exchangers

I_{NaK}. A plausible model for I_{NaK} consists of a pump gain modified by a Michaelis-Menten kinetic dependence on internal Na^+ and external K^+ concentrations (Cooke et al. 1974; DiPolo 1979; Mullins 1981). Because neither of these concentrations vary in our model, a constant value was assumed for the Na^+ pump current I_{NaK}. In an endogenously bursting cell, if I_{NaK} is blocked by ouabain, a beating mode ensues (Junge and Stephens 1973). A value on the order of a few nanoamperes was assumed for I_{NaK} on the basis of the amount of current required to bias the operation of the cell from a bursting to a beating mode (Arvanitaki and Chalazonitis 1968), or equivalently, from a slowly oscillating mode to a constant depolarized mode (Futamachi and Smith 1982; Mathieu and Roberge 1971) (see RESULTS).

I_{NaCa}. We achieved a Ca^{2+} balance in the model by employing a Na^+-Ca^{2+} exchanger in combination with a Ca^{2+} pump. Such an electrogenic Na^+-Ca^{2+} exchanger has been shown to be operational in squid axon with a probable ratio of three to five Na^+ ions to one Ca^{2+} ion (Baker et al. 1969; Blaustein and Hodgkin 1969). More recent studies have demonstrated the presence of a Na^+-Ca^{2+} exchanger in the somata of *Aplysia* neurons (Satin 1984; Zucker 1989).

The general form of the equation (APPENDIX, *Pumps and exchangers*) for I_{NaCa} is the same as the one used by DiFrancesco and Noble (1985) and Rasmusson et al. (1990a,b) in their models of cardiac activity [see Mullins (1977, 1981) for derivation, and see DiFrancesco and Noble (1985) for an explanation of the various parameters]. The major assumption is that the only energy sources available to the exchanger are the concentration gradients. At a given membrane voltage and Ca^{2+} concentration, the exchanger generates a net current. For example, at a 4:1 ratio, the four Na^+ ions have a combined charge of 4 electrostatic units (esu), and the lone Ca^{2+} ion has a charge of 2 esu resulting in an imbalance of 2 esu for each Ca^{2+} ion exchanged. During Ca^{2+} efflux, the exchanger generates a net inward current, and during Ca^{2+} influx, an outward current. The net flux is the difference between influx and efflux of Ca^{2+}. Because Ca^{2+} influx generates a net outward movement of positive charge, it is favored by the membrane field when the membrane potential is depolarized (Mullins 1977). Thus the driving force for Ca^{2+} influx varies as $e^{VF/RT}$, and the influx dominates at depolarized potentials. The concentration dependence of the Ca^{2+} influx is given only in terms of $[Ca]_o$ and $[Na]_i$. Conversely, the driving force for Ca^{2+} efflux varies as $e^{-VF/RT}$ and dominates at hyperpolarized potentials; its concentration dependence is given in terms of $[Ca]_i$ and $[Na]_o$. Thus I_{NaCa} is both concentration dependent and voltage dependent. Because $[Ca]_o$, $[Na]_i$, and $[Na]_o$ are held constant in this model, I_{NaCa} is essentially dependent on $[Ca]_i$ and membrane voltage (V).

Two factors influenced the parameter selection for this current: *1*) the dependence of the reversal potential of I_{NaCa} on the flux ratio (r) and *2*) the tendency of this current to negate the region of negative resistance introduced by I_{SI}, which is essential for bursting. Because the temperature and three of the four applicable concentrations are fixed in the model, the reversal potential of this current depends only on $[Ca]_i$ and the flux ratio. A flux ratio of 4:1 seems plausible on the basis of data from squid axons (Requena 1983) and was selected because the steady-state *I-V* relationship that it generated produced the most accurate simulation of bursting activity.

I_{CaP}. A high-affinity, low-capacity, electrogenic Ca^{2+} extrusion pump, which generates an outward current (I_{CaP}) that increases with increasing Ca^{2+} concentration, has been described in squid axon (DiPolo and Beaugé 1979; Requena 1983). The Ca^{2+} efflux associated with the ATP-driven pump is referred to as "uncoupled" to differentiate it from that associated with the Na^+-Ca^{2+} exchanger. A half-activation of 180 nM and a maximal rate of 0.15 pmol cm^{-2} s^{-1} have been reported for squid axon (DiPolo and Beaugé 1979). Nearly all of the Ca^{2+} efflux observed at physiological concentrations of Ca^{2+} was due to the uncoupled Ca^{2+} efflux, whereas the Na^+-Ca^{2+} exchanger was more significant at very high concentrations. The maximal pump rate cited above for squid axon would produce a current of only 0.5 nA in R15; this rate was not sufficient to cause the equilibrium point of the model to fall in a negative slope resistance region of the *I-V* curve (see RESULTS). It has been suggested that the maximal pump rates have been underestimated in cases where there is significant cytoplasmic binding of Ca^{2+} (Stockbridge and Moore 1984). The gain and half-activation level of the Ca^{2+} pump were adjusted to produce a greater efflux of Ca^{2+} (APPENDIX, *Pumps and exchangers*). The maximum current generated by the pump in any simulation was <4 nA.

Calcium regulation

The model includes a description of the fluid compartments associated with the distribution of ions including an internal buffering compartment for Ca^{2+} uptake and release (Fig. 3). The effective cell volume was set to 4.0 nl, which includes a factor of one-half to account for the fact that some of the cell volume is occupied by Ca^{2+} buffers. Several studies have shown that <1% of

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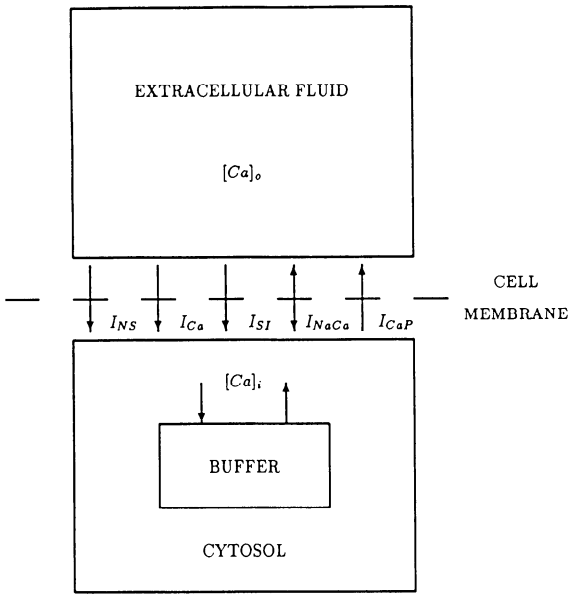


FIG. 3. Compartmental model for calcium balance. Calcium can accumulate in 2 intracellular compartments: the cell cytosol and on the buffer. The Ca^{2+} concentration in the well-stirred extracellular fluid compartment is assumed to be constant.

the calcium that enters the cell during a pulse of I_{Ca} remains unbuffered for any measurable length of time (Gorman et al. 1984; Plant et al. 1983). In the absence of precise knowledge of the type of cytosolic buffer in R15, we assumed that the buffer had properties similar to the Ca^{2+} binding properties of calmodulin (Robertson et al. 1981). The rate of change of internal free Ca^{2+} concentration ($[Ca]_i$) is described by a first-order differential equation of the form

$$[\dot{Ca}]_i = \frac{\sum w_j I_j}{Vol_i F} - n[B]_i O_C \quad (4)$$

where I_j is the net current (nA), w_j is the ratio of net current in electrostatic units (esu) per second to the net ionic flow of Ca^{2+} given in ions per second, Vol_i is effective cell volume (nl), F is Faraday's constant, $[B]_i$ is the internal concentration of cytosolic buffer (μM), n is the number of binding sites on the buffer, and O_C is the occupancy (fraction of the sites already occupied by Ca^{2+} ions and therefore unavailable for calcium binding). Calcium binding to the intracellular buffer is assumed to be a first-order process described by the following differential equation for the time rate of change of occupancy of calcium binding sites (Robertson et al. 1981)

$$\dot{O}_C = k_U[Ca]_i(1 - O_C) - k_R O_C \quad (5)$$

where k_U is the first-order kinetic rate constant ($mM^{-1} \cdot ms^{-1}$) for the uptake of Ca^{2+} and k_R is the rate constant (ms^{-1}) for its release (see APPENDIX *Internal calcium concentration*). Our equations for the intracellular Ca^{2+} balance are analogous to those given by Rasmusson et al. (1990a,b) for the cardiac myocyte.

In the reduced two-variable model described in the RESULTS, we assumed that rapid equilibrium was achieved between free calcium ($[Ca]_i$) and buffered calcium ($n[B]O_C$). We then scaled the derivative of $[Ca]_i$ (APPENDIX, *Internal calcium concentration*) by the ratio of free to total Ca^{2+} (Chay and Keizer 1983) with O_C set to its steady-state value.

Model implementation

The model has 11 state variables, including membrane potential, internal Ca^{2+} concentration, Ca^{2+} buffer occupancy, and 8 membrane gating variables. Steady-state activation and inactivation curves for the gating variables associated with the ion channel currents are plotted in Fig. 2 as a function of membrane voltage. Typical voltage-clamp simulations showing the magnitude and dynamics of some of the ionic membrane currents are shown in Fig. 4. Note that both fast and slow currents are shown in this figure, and the time scales are different in each panel. To simulate a voltage-clamp experiment, the derivative of the membrane potential, $\dot{V}$ in *Eq. 1*, was set to zero, and the model equations were solved for specific currents before, during, and after step changes in membrane potential. Internal Ca^{2+} concentration was initialized at its steady-state value at the holding potential, and the Ca^{2+} buffer occupancy was set to its equilibrium value for that Ca^{2+} concentration.

An implicit Runge-Kutta method of order five with variable step size (Hairer and Wanner 1990) was used to solve this stiff system of differential equations on a Sequent Symmetry S27 multiprocessor computer. Typical run time using a single processor was ~12 min for 1 min of simulation in which bursting occurred. This algorithm achieved a significant reduction in run time as compared with a fourth-order explicit Runge-Kutta method with variable step size.

RESULTS

The model simulates each of the general aspects of the activity of R15 as described in the INTRODUCTION and provides insights into the particular mechanisms underlying these features. We examined three closely related variants of the R15 model. The model described in the METHODS and comprised of the equations and parameters given in the APPENDIX will be denoted model Ia. This complete model, with 11 state variables, exhibited bursting, beating, and silent modes of activity. In version Ib, a single parameter change was made: $\bar{g}_{Na}$ was set to zero, effectively uncoupling the slow dynamics of the model from the fast-spiking dynamics and reducing the model order to nine. The slow dynamics, which we then observed in isolation, ranged from a silent hyperpolarized state to an oscillatory "slow-wave" mode to a constant depolarized level. A reduced, two-variable model (II) was also examined. The order of the original model was reduced by setting all of the gating variables to their steady-state values (z_∞) and the buffer occupancy to its equilibrium value; the two remaining variables were intracellular Ca^{2+} and membrane potential. As in model Ib, $\bar{g}_{Na}$ was set to zero. All simulations (with the exception of the dashed-line *insets* in Fig. 7) were generated with either model Ia or Ib. On the other hand, all phase-plane and *I-V* relationship analyses were generated by the use of model II. We employed model II to predict the existence of slow-wave activity in model Ib and, to some extent, the full dynamics exhibited by model Ia. In addition, several parameter adjustments were made to the complete model (Ia) to demonstrate the predictive capability of the model with respect to the effects of modulatory agents.

Complete model (Ia)

Injected current (I_{STIM}) was used to manipulate the interburst potential level and hence the mode of activity. Simulations of endogenous bursting activity, the hyperpolarized (silent) mode, and the beating mode are shown in Fig. 5*A*, as well as the transitions from the bursting mode (no in-

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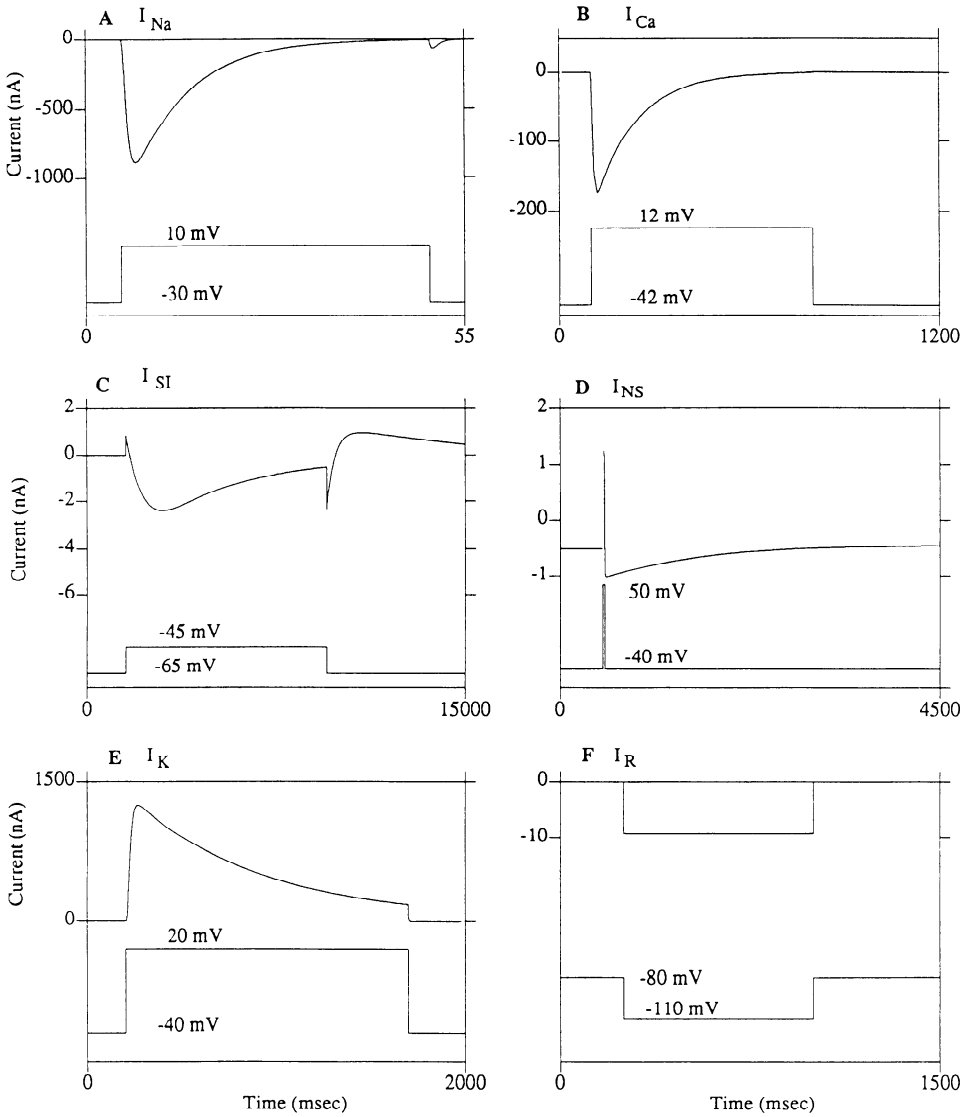


FIG. 4. Typical voltage-clamp simulations of key currents in the model. Note that the time and potential axes vary widely in scale. Voltage-clamp simulations are based on actual experimental protocols (Adams and Benson 1985; Adams and Gage 1979a; Adams and Levitan 1985; Hermann and Gorman 1981a,b). The kinetics of the currents are somewhat different from published data, because they have been scaled to a temperature of 22°C.

jected current) to the beating mode (depolarizing stimulus current) and the silent mode (hyperpolarizing stimulus current). Values of I_{STIM} were chosen to illustrate the range of the dynamic activity of the model. The nature of the transitions is qualitatively very similar to that observed experimentally in R15 (Arvanitaki and Chalazonitis 1968). As the injected current varies from several nanoamperes of hyperpolarizing current to several nanoamperes of depolarizing current, the cell spends a larger fraction of time spiking, i.e., the number of spikes per burst increases, and the interburst interval decreases. To understand how the model simulates these transitional features, it is necessary to examine the slow dynamics of the system and particularly the slow oscillation in potential underlying the burst activity. This subject is addressed in the following section. An apparently chaotic transitional mode was observed (not shown) between the bursting and beating modes as I_{STIM} was varied. Other transitions between modes, induced by changing parameters such as g_{SI} or $\bar{g}_{NS}$, were also effective in producing apparently chaotic waveforms. Similar transitional modes have been observed both experimentally (Smith and Thompson 1987) and in previous models (Canavier et al. 1990).

Waveforms of the intracellular concentration of Ca^{2+} ($[Ca]_i$) associated with the various modes of electrical activity are shown in Fig. 5*B*. In the silent mode the calcium concentration, like the potential, is constant. However, in the simulations of bursting activity, a slow variation in $[Ca]_i$ is observed. This variation has the same period as the slow variation in potential on which the bursts are superimposed. The magnitude and time course of the simulated changes in $[Ca]_i$ are in reasonable agreement with the available data (Gorman et al. 1981, 1984; Gorman and Thomas 1978). In the beating mode the variation in $[Ca]_i$ is relatively low in amplitude and has the same period as the rapid oscillations in potential.

The model provides the following explanation of bursting activity. When the potential of the slow oscillation is above the spike threshold for the cell, the cell discharges action potentials, otherwise it is silent. The slow inward current, I_{SI}, together with the mechanisms for Ca^{2+} regulation, is responsible for the slow cyclic variation in both Ca^{2+} concentration and potential. If $\bar{g}_{SI}$ is set to zero, bursting activity is abolished, irrespective of the value of I_{STIM}. During the hyperpolarized interval between the crests of the slow wave, Ca^{2+} is removed from the cell (Fig. 5*B*), so the

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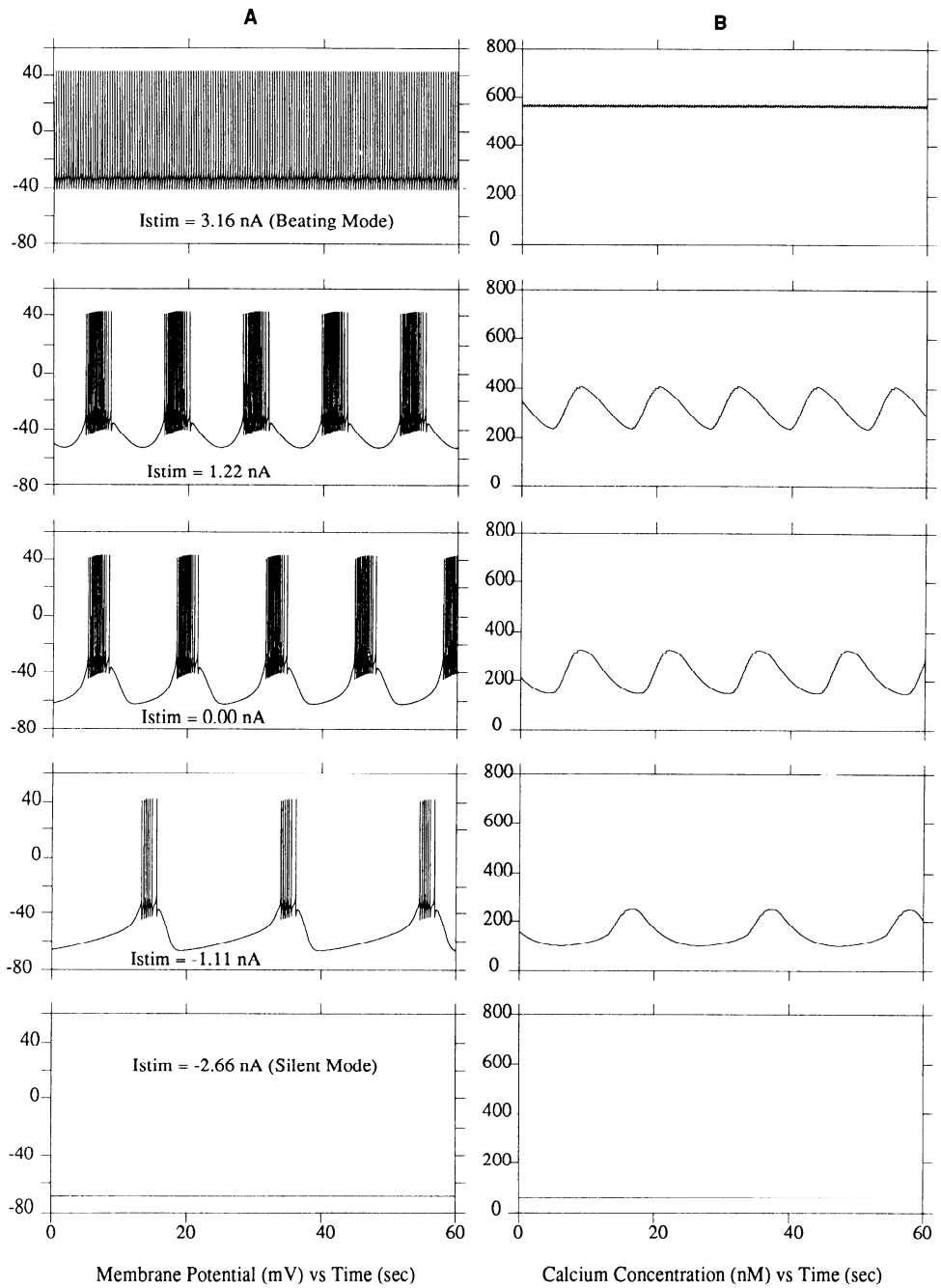


FIG. 5. Simulation of different modes of activity as a function of applied current. *A*: membrane potential. *B*: calcium concentration. With hyperpolarizing current (−2.66 nA) the cell is silent, but as the hyperpolarization is reduced, the model generates a bursting rhythm with the frequency, interburst interval, and magnitude of the interburst trough being a function of the applied current. As the depolarizing current increases (3.16 nA), the simulated activity switches from a bursting to a beating mode. Note that a stimulus current of 3.16 nA exactly cancels the current generated by the Na^+-K^+ pump (I_{NaK}).

Ca^{2+} inactivation of I_{SI} is gradually removed. As the cell then depolarizes, I_{SI} is regeneratively voltage activated. When the slow wave is near its crest, Ca^{2+} accumulates in the cell, gradually inactivating I_{SI}. The cell then hyperpolarizes, I_{SI} is regeneratively voltage deactivated, and the cycle begins once again.

Reduced model

SLOW DYNAMICS. As indicated by the simulations of Fig. 5, the existence of a slow cyclic variation in $[Ca]_i$ and membrane potential is a key determinant of the activity of R15. In the following section we show that insights into the modes of activity of R15 can be obtained by reducing the model to two dimensions and examining the system dynamics in terms of two variables, $[Ca]_i$ and membrane potential (V). Such a reduced system (i.e., model II) is amenable to a phase-plane analysis, wherein the mode of activity is determined by an analysis of the stability of the fixed points (also called equilibium or singular points) occurring at the intersections of the potential and calcium nullclines (for reviews of phase-plane analysis, see Fitzhugh 1969; Rinzel and Ermentrout 1989; Segel 1980; Thompson and Stewart 1986). First, we will describe how the fixed points are located. Second, we will show how to determine their stability (with respect to the slow dynamics) and thus predict the existence of slow oscillations (model Ib) that determine the mode of activity of the cell. The location of the fixed point as determined for the reduced system is also exact for the nine-dimensional system (Ib). However, the stability analysis is only approximate because of the assumption that the seven omitted variables are fast com-

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pared with the remaining two. Indeed, the gating variables *b, l*, and *s* can vary slowly under certain conditions. Thus this analysis only applies for a range of parameters. [For a more complete mathematical treatment of the slow variables, including a reduced $s - [Ca]_i$ model, see Rinzel and Lee (1987).] Third, we will apply the predictions of the reduced system to the full model (Ia), including bursting activity.

PHASE-PLANE ANALYSIS. A nullcline is comprised of the points at which the time rate of change (1st derivative) of the variable is zero. Thus the calcium nullcline for the reduced model II as seen in Fig. 6 (thick dashed curve) was generated by solving the equation for $[\dot{C}a]_i$ in the APPENDIX, *Internal calcium concentration*, implicitly for the potential required for $[\dot{C}a]_i$ to equal zero at a given $[Ca]_i$ (with each current set to its steady-state value and the derivative of the buffer occupancy ($\dot{O}_C$) set to 0). Because this equation defines the calcium nullcline implicitly rather than explicitly, the solution was implemented by trial and error with the use of a computer. The primary currents that determine the calcium nullcline are I_{SI}, I_{NaCa}, and I_{CaP} (I_{Ca} and I_{NS} do not contribute significantly at potentials more negative than −20 mV). As $[Ca]_i$ is increased *1)* I_{CaP} increases the efflux of Ca^{2+} from the cell, and *2)* I_{SI} inactivates. Therefore the potential must become more depolarized to produce a net increase in Ca^{2+} influx via I_{SI} to offset the additional Ca^{2+} efflux. Synergistically, an increase in potential drives I_{NaCa} toward increased Ca^{2+} influx. At potentials more depolarized than about −20 mV, the nullcline bends to the right, reflecting the large influx of Ca^{2+} at depolarized potentials because of the activation of I_{NS} and I_{Ca}.

The potential nullcline was generated in a similar manner by solving *Eq. 1* implicitly for the value of $[Ca]_i$ that results in $\dot{V}$ equal to zero at each value of V (with each current set to its steady-state value). The implicit dependence of the potential nullcline on calcium necessitated the use of a trial-and-error solution in this case as well. Effectively, the sum of the ionic membrane currents is held constant (and equal to I_{STIM}) as the potential is varied. The "Z" shape of the potential nullclines can be explained as follows (see, for example, the curve in Fig. 6 parameterized by an I_{STIM} of −2.66 nA). As the potential is varied from −10 to

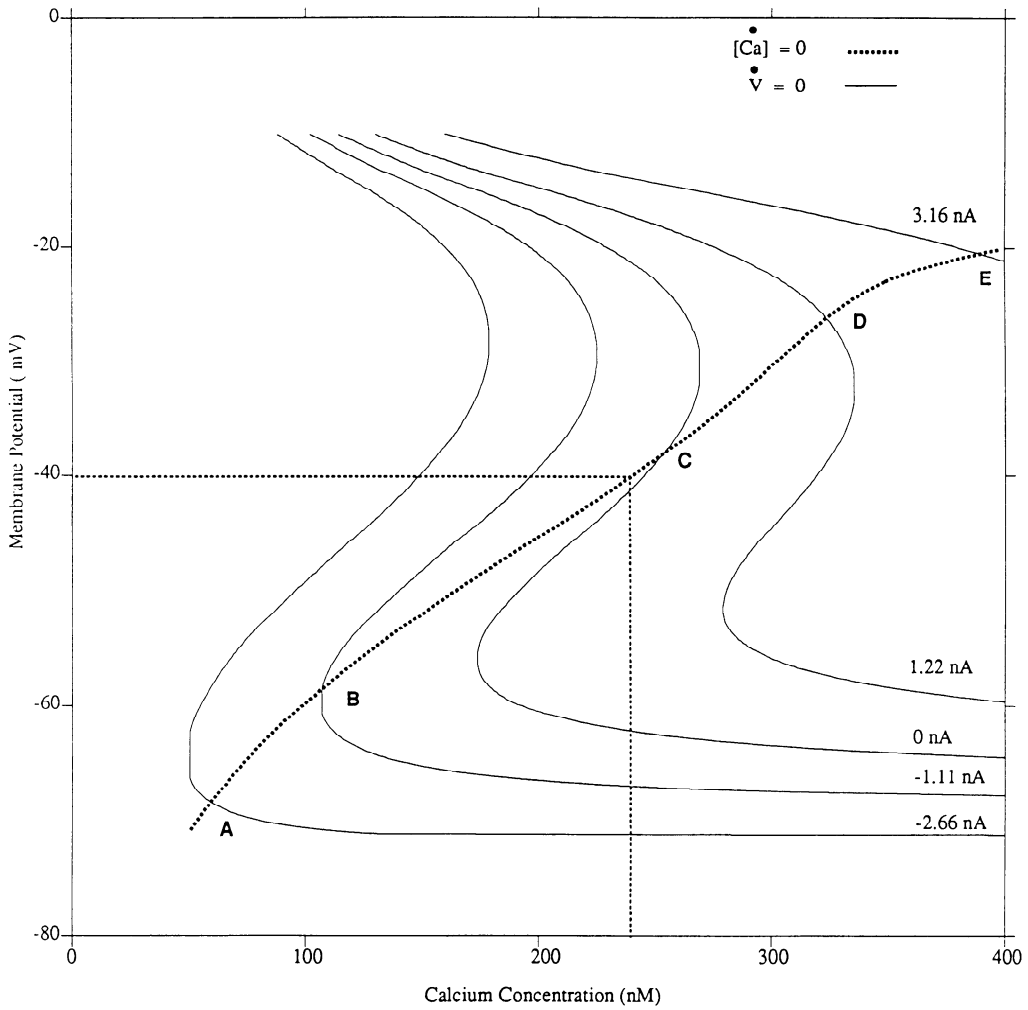


FIG. 6. Nullclines for the reduced system parameterized by I_{STIM}. In the reduced system (II), $\bar{g}_{Na}$ is set to 0, all ionic currents are set to their steady-state values, and $\dot{O}_C$ is set to 0. The calcium nullcline was calculated by implicitly solving for the value of membrane potential for which $[\dot{C}a]_i$ is equal to 0 for each value of $[Ca]_i$ (see the 1st equation in the APPENDIX, *Internal calcium concentration*). An example calculation is shown to obtain the potential (−40 mV) required for Ca^{2+} equilibrium at $[Ca]_i$ = 240 nM. The potential nullclines were calculated by implicitly solving for the value of $[Ca]_i$ for which $\dot{V}$ in *Eq. 1* is equal to 0 at each value of V. The potential nullclines are parameterized by external stimulus current.

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−72 mV, the sum of the purely voltage-dependent currents (primarily I_K, I_L, and I_R) becomes more negative (less outward). Because $[Ca]_i$ is the only variable in this calculation, $[Ca]_i$ must increase or decrease as appropriate to balance the voltage-dependent decrease in outward current with a Ca^{2+}-dependent increase in outward current. We observed the following. *1*) In the upper branch of the Z (from approximately −10 to −28 mV at I_{STIM} = −2.66 nA), $[Ca]_i$ increases from 100 to 180 nM, which augments the outward current I_{CaP} and inactivates the inward current I_{SI}. *2*) In the middle branch of the Z (approximately −28 to −64 mV), the deactivation of I_{SI} with more hyperpolarized potentials becomes significant. An inflection point occurs when a further increase in $[Ca]_i$ would overcompensate for the voltage-dependent reduction in outward currents. Thus a decrease in $[Ca]_i$ from 180 to 50 nM is associated with this part of the nullcline. *3*) Finally, in the lower branch of the Z (approximately −64 to −72 mV), I_{SI} is deactivated to the point that its reduction no longer dominates the nullcline computation, and $[Ca]_i$ increases, which, in turn, increases the outward contribution from I_{CaP}. This lower branch is flat because I_{NaCa} increases rapidly at more hyperpolarized potentials so that a steep increase in both $[Ca]_i$ and I_{CaP} is required.

A positive change in I_{STIM} causes the potential nullcline to be shifted to the right (higher $[Ca]_i$) and causes the middle branch to be shortened, which, in turn, shifts the lower branch upward. Again, the explanation can be given in terms of the calcium-dependent currents. As I_{STIM} is made more positive, more outward current is required at a fixed potential, so a higher $[Ca]_i$ results because of a need for increased I_{CaP} and decreased I_{SI}. At higher $[Ca]_i$, I_{SI} is smaller and so is the middle branch of the Z for which it is responsible.

The calcium nullcline is in reasonable agreement with the steady-state calcium concentration levels experimentally measured (relative to the resting concentration at −50 mV) during voltage-clamp steps lasting 20 s (Gorman and Hermann 1982). The various levels of stimulus current employed in Fig. 6 correspond with those shown in Fig. 5. The intersection of the two nullcline curves in the phase plane generated by V and $[Ca]_i$ is the fixed point of the system with the corresponding value of I_{STIM}. A concept central to the analysis of this system is that any change in parameters changes the location of the fixed point. This concept is illustrated by the rather dramatic effects that small changes in I_{STIM} have on the location of the fixed point (A–E in Fig. 6).

If one assumes that $[Ca]_i$ varies slowly with respect to V, the slope of the potential nullcline at the fixed point determines whether a fixed point is stable with respect to the slow dynamics of the model. If the slope is negative, then any perturbation along the nullcline tends to return the system to the fixed point; therefore the point is stable and thus classified as an attractor. If the slope is positive, however, the opposite is true: the fixed point is unstable and repels all nearby orbits to a limit cycle. A limit cycle in the phase plane corresponds to a periodic solution of the system equations (Segel 1980; Thompson and Stewart 1986).

The nullcline analysis of the reduced two-variable system (II) correctly predicted the dynamic behavior of the nine-state variable system (Ib). For example, the fixed points B and C are associated with positive slopes on the potential nullclines in Fig. 6, hence a slow oscillation was observed when the nine-variable system (Ib) was solved at the corresponding parameter values (see *inset*, Fig. 7). For comparison, the oscillatory solutions to the two-variable system for the same values of I_{STIM} are also shown in the *insets* to Fig. 7; the instantaneous behavior of the gating variable s causes the frequency to be higher in the reduced system. On the other hand, points A, D, and E in Fig. 6 (−2.66, 1.22, and 3.16 nA, respectively) are associated with a negative slope, and thus this analysis predicted a stable state, at a hyperpolarized level in the case of point A and at a depolarized level in the case of points D and E. Simulations also confirmed the predictions at points A, D, and E (not shown). In the section on application of the reduced model to the complete dynamics, we will explain the consequences of adding the Na^+ current back into the model.

STEADY-STATE *I-V* RELATIONSHIPS (MODEL II). Although the analysis in the previous section establishes a useful theoretical framework for describing the electrical activity of bursting neurons, most physiological analyses focus on the characteristics of the steady-state *I-V* relationship. Previous theoretical analyses of this relationship (Benson and Adams 1987; Hindmarsh and Rose 1984) have several features in common with the phase-plane analysis of the preceding section: *1*) the gating variables were assumed to have reached steady state (effectively reducing the system to a single dimension, membrane potential, V), *2*) the criterion of $\dot{V} = 0$ was used to locate the fixed points, and *3*) the slope of a curve (in this case the steady-state *I-V* relationship) passing through the fixed point was used to determine its stability. Because our reduced model (II) contains an additional state variable ($[Ca]_i$), and several currents depend on $[Ca]_i$, there is no single analogous steady-state *I-V* relationship for the model, but rather a continuous family of such curves parameterized by $[Ca]_i$ (see Fig. 7) (Rinzel 1987). This generalization can be conceptually justified by assuming that, on the time scale relevant to the determination of the stability of a fixed point with respect to the slow dynamics, the variation in $[Ca]_i$ is slow relative to that of the gating variables, because of the dampening effect of the cell volume and the buffering of intracellular Ca^{2+}.

The *I-V* relationships in Fig. 7 were generated by fixing $[Ca]_i$ and computing the sum of the steady-state ionic membrane currents (I_{SS}) as a function of V. This sum also can be interpreted as the stimulus current level required for the total current and hence $\dot{V}$ to equal zero (see *Eq. 1*). The calculation of the *I-V* relationship is the inverse of the one used to generate the potential nullclines in Fig. 6, so it is not surprising that they contain similar information about stability. The *I-V* relationships thus generated are roughly N shaped (Gola 1974; Wilson 1982; Wilson and Wachtel 1974); the leftmost region of positive slope is primarily due to I_R, the region of negative slope to I_{SI}, and the rightmost region of positive slope to I_K (see also Benson and Adams 1987). Figure 7 illustrates five *I-V* relationships generated at five different levels of $[Ca]_i$. These levels of $[Ca]_i$ were selected at the intersections of the calcium nullcline in Fig. 6 with the potential nullclines parameterized by the five

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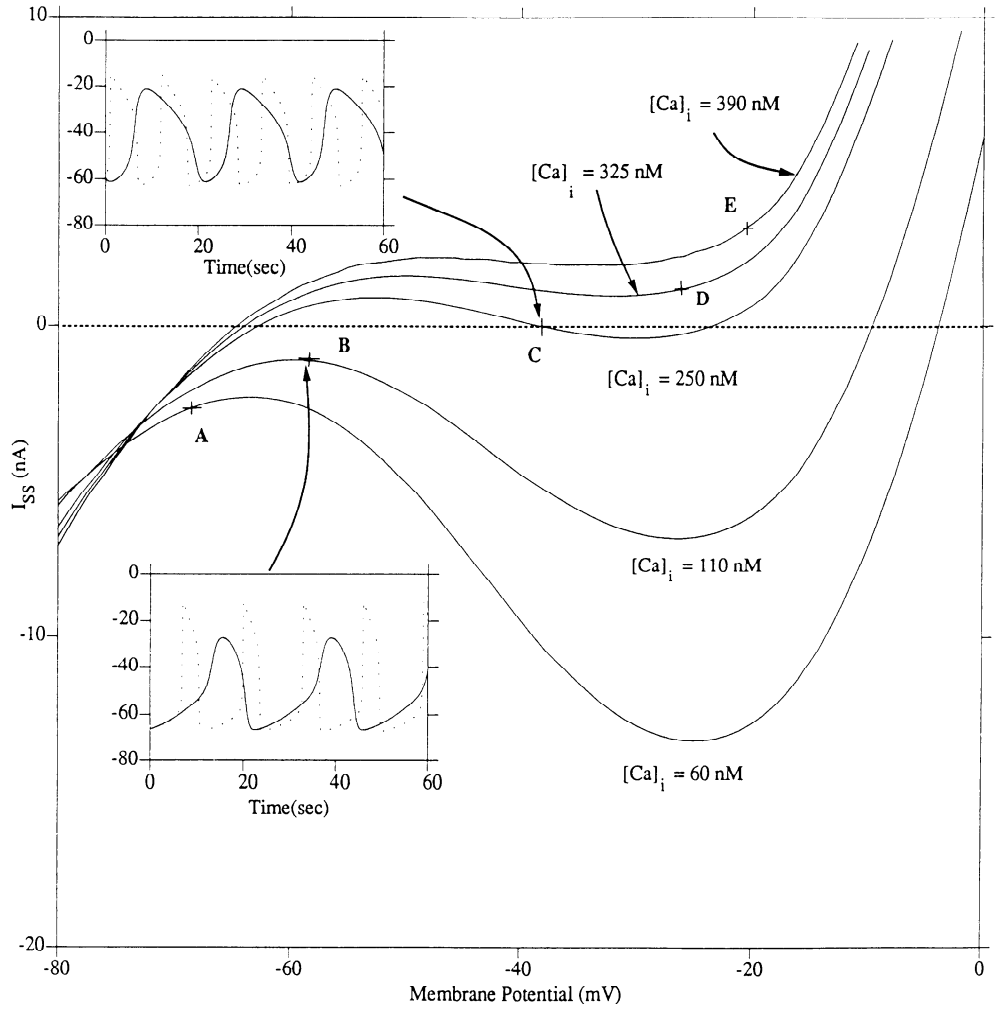


FIG. 7. Steady-state *I-V* curve. Current was calculated with all gating variables set to their steady-state values and $\bar{g}_{Na}$ set to 0. Crosses are positioned at points where $[\dot{C}a]_i = 0$ for the corresponding values of V and $[Ca]_i$, as determined from Fig. 6. The membrane potential, steady-state current level, and Ca^{2+} concentration associated with each of the labeled points are as follows: A: −68 mV, −2.66 nA, 60 nM; B: −58 mV, −1.11 nA, 110 nM; C: −38 mV, 0.0 nA, 250 nM; D: −26 mV, 1.22 nA, 325 nM; and E: −21 mV, 3.16 nA, 390 nM. *Insets* (solid lines) show slow-wave activity associated with model Ib biased at points B and C, with I_{STIM} set to −1.11 nA and 0 nA, respectively. Dashed lines in the *insets* show the waveforms generated by the reduced 2-variable system.

different values of I_{STIM}. For example, the *I-V* curve containing the cross symbol labeled by the C was generated with a $[Ca]_i$ of 250 nM. The crosses in Fig. 7 indicate that $[\dot{C}a]_i = 0$ at the corresponding values of $[Ca]_i$ and V. For example, the cross labeled with the C was positioned on the *I-V* curve at the membrane potential (−38 mV) corresponding to the intersection of the nullclines at point C in Fig. 6. Recall that in Fig. 6 a fixed point occurs at the intersection of the nullclines for calcium ($[\dot{C}a]_i = 0$) and potential ($\dot{V} = 0$). For $\dot{V}$ at a point to equal zero, I_{SS} must equal the parameter value of I_{STIM} for the model (see *Eq. 1*). Therefore the fixed point of the system is represented by a unique point (located on a unique *I-V* relationship parameterized by a unique $[Ca]_i$) at which the I_{SS} coordinate of the point labeled with a cross equals I_{STIM}. Thus the fixed point with $I_{STIM} = 0$ corresponds with point C in Fig. 7. The slope of the steady-state *I-V* curve at the intersection with the system equilibrium point determines whether the point is stable. An analysis of the effect of small perturbations reveals that, in this case, a positive slope is stable, whereas, in the case of the nullclines, the opposite was true.

The behavior of the system can be altered by biasing the system so that the equilibrium point has the desired characteristics. The advantage of using the *I-V* curves over the nullclines is that the effect of changing a specific current can be readily visualized. For example, the location of the fixed point in Fig. 7 can be shifted by applying the appropriate stimulus current (I_{STIM}). This is analogous to experimental protocols such as injecting a constant hyperpolarizing or depolarizing current, or blocking the Na^+-K^+ pump. The predictions based on the *I-V* relationship agree with those based on the phase-plane analysis of the preceding section. For example, to mimic constant hyperpolarization, the system should be biased (with a hyperpolarizing current of 2.66 nA) so that there is zero total current at point A, where the slope is positive resulting in a stable fixed point. The slope of the steady-state *I-V* curve is negative at points B and C, and therefore a subthreshold oscillation is predicted. Simulations employing the nine-variable model Ib confirm this prediction and show how the character of the slow-wave oscillation is changed as the operating point progresses from B to C, with bias currents of −1.11 and 0 nA,

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respectively (see *insets* in Fig. 7). To produce a constant depolarization, the system should be biased (with a depolarizing current of 1.22 and 3.16 nA, respectively) at points D and E, where the slope is again positive.

APPLICATION OF REDUCED MODEL (II) TO COMPLETE DYNAMICS (Ia). The preceding two sections are concerned solely with the prediction of slow-wave activity in model Ib ($\bar{g}_{Na}$ set to zero). To extend the predictions to the full model (Ia), we assume that, whenever the slow potential exceeds the spike threshold (about −25 mV), spiking activity will occur. Like the assumption that $[Ca]_i$ varies slowly with respect to the gating variables, this assumption is only valid in a range of initial conditions and parameters (see Rinzel and Lee 1987). Further, when extrapolating the analysis of model II to the full model Ia, one must take into account the change in the nullclines and steady-state *I-V* curves at depolarized potentials, because of a nonzero $\bar{g}_{Na}$. Qualitatively (at least in terms of the stability of the fixed point), the predictions of the reduced model with $\bar{g}_{Na} = 30$ μS agree with those for $\bar{g}_{Na} = 0$. Thus the analysis of the reduced model (II) correctly predicts the modes observed in Fig. 5 and generated by model Ia, namely a hyperpolarized state at point A (stable fixed point below the spike threshold), bursting at points B and C (unstable fixed point), and beating at point E (stable fixed point above the beating threshold). However, the reduced model predicts beating activity when the model is biased at point D, when in fact bursting activity was observed. The slow underlying oscillation that is not predicted by the reduced $V - [Ca]_i$ model may be associated with the slow variable b, because if $\bar{g}_{NS}$ is set to zero, then beating activity is actually generated as predicted when the model is biased at point D. Thus the fast and slow dynamics interact in ways that cannot be predicted by the slow $V - [Ca]_i$ dynamics alone. For example, each spike admits additional calcium into the cell and activates I_{NS}, which, in turn, boosts spiking frequency. Hence the frequency of the bursts (see Fig. 5 at $I_{STIM} = -1.11$ and 0 nA) does not exactly correspond to the frequency of the slow potential oscillations (see Fig. 7, *insets* at points *B* and *C*) for the same value of I_{STIM}. The qualitative nature of the change in shape of the underlying slow potential oscillation is preserved, however.

Burst characteristics

Next we examined the characteristics of the simulated bursts. The model currents during a typical burst, which are not directly measurable, are shown in Fig. 8. As indicated on Figs. 5 and 8, there is a decrease in hyperpolarizing afterpotential, a parabolic variation in interspike interval, and the presence of the DAP. In addition, there is a small increase in overshoot (which is not obvious with these potential scales).

The parabolic bursting pattern results at least partially from two competing processes, the slow voltage-dependent activation of I_{SI} and the even slower calcium-dependent inactivation of the same current, which determine the character of the slow wave. The general shape of the underlying slow wave in R15 has been referred to as sinusoidal as opposed to square such as that observed in pancreatic β cells (Chay and Cook 1988; Chay 1990). The character of the slow wave can be adjusted from square to sinusoidal by increasing the value of τ_s, the time constant for the activation variable for the slow inward current (I_{SI}). Bursts that are associated with a sinusoidal slow wave tend to produce spikes that increase in amplitude and frequency until the crest of the slow wave, then decrease in amplitude and frequency, producing the characteristic parabolic bursting pattern (Chay and Cook 1988; Chay 1990; Rinzel and Lee 1987; Rinzel 1987). The R15 cell deviates from this general description only in that the maximum height of the spikes does not generally decrease when the frequency starts to decrease but, instead, remains constant. In our simulations, the decrease in undershoot was due to cumulative inactivation of I_K.

We found that I_{NS} was not necessary to produce any of the characteristics of a burst described above. However, without I_{NS}, the DAP lasted several seconds longer in some cases, there were fewer spikes per burst, and the spiking frequency did not increase as rapidly during the initial part of a burst. In addition, simulations that included a description of the transient outward current (I_A) showed that the character of the burst was not altered by its inclusion. Thus a multiplicity of currents is not necessary to account for the characteristics of a burst.

Effects of modulatory agents

The electrical activity of R15 is profoundly affected by a variety of neurotransmitters including serotonin (5-HT), egg-laying hormone (ELH), dopamine, and FMRFamide. Model Ia was tested further by examining its ability to simulate the known effects of these agents. The effects of 5-HT and ELH are believed to be due to an adenosine 3′,5′-cyclic monophosphate (cAMP)–mediated enhancement of I_R and, in some cases, an enhancement of I_{SI}. On the other hand, dopamine and FMRFamide are believed to produce their effects via a cyclic guanosine monophosphate (cGMP)–mediated reduction of I_{SI}.

Dopamine and FMRFamide induce a silent hyperpolarized state in a normally bursting R15 (Benson and Adams 1987; Kramer et al. 1988; Lotshaw and Levitan 1988). In contrast, the application of cGMP analogue can convert a bursting neuron into a slow beater (Levitan and Levitan 1988a). This result is surprising because all three agents are believed to produce their effects via a reduction in I_{SI} (Kramer et al. 1988; Levitan and Levitan 1988a).

5-HT at low concentrations drives the activity of the cell toward a more hyperpolarized state, frequently rendering the cell silent (Benson and Adams 1987). The enhancement of I_R at low concentrations of 5-HT is believed to be responsible for this effect (Adams and Benson 1985; Benson and Adams 1987; Benson and Levitan 1983; Levitan and Levitan 1988b; Lotshaw and Levitan 1988). Although either dopamine or 5-HT can induce a hyperpolarized state, a cell that has been exposed to dopamine cannot be induced to burst by applying a depolarizing current, whereas a cell hyperpolarized by 5-HT can (Lotshaw and Levitan 1988). Both ELH and high concentrations of 5-HT augment the depolarizing phase of the bursting activity as well as the hyperpolarizing phase (Levitan and Levitan 1988b; Mayeri et al. 1979). ELH and high concentrations

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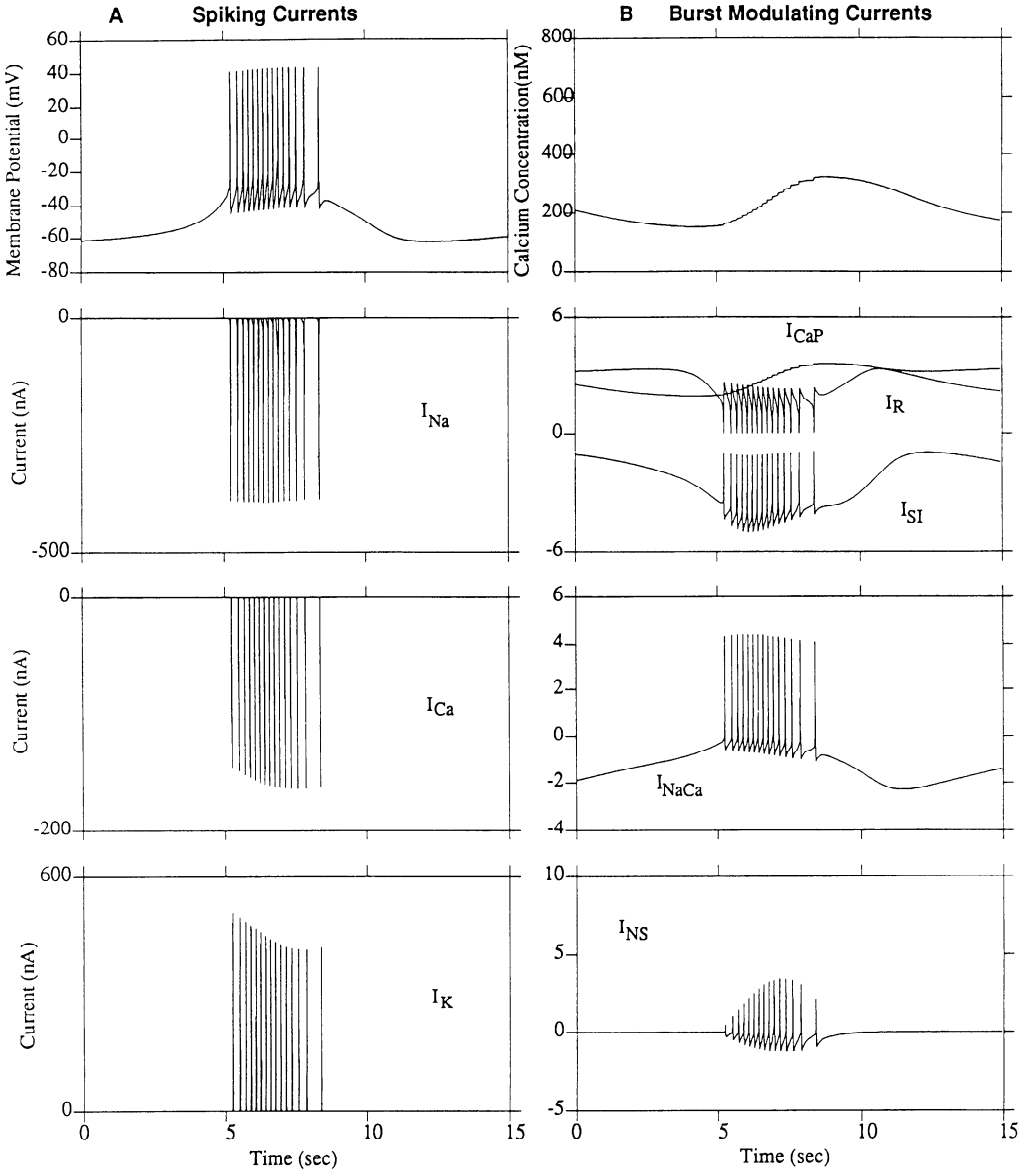


FIG. 8. Membrane currents contributing to the electrical activity. *A: top panel* illustrates a burst generated with the same parameters as Fig. 5 (I_{STIM} = 0) but shown on an expanded time scale. The large-amplitude currents contributing to the spikes during a burst are plotted below the membrane potential on the same time scale. *B*: corresponding levels of $[Ca]_i$ are shown in the *top panel*, and the low-amplitude currents contributing to the slow oscillations of potential are illustrated below.

of 5-HT are known to enhance both I_R and I_{SI}, and these opposing effects are believed to account for augmented bursting (Kramer and Levitan 1990; Levitan and Levitan 1988b; Mayeri et al. 1979).

The model was able to simulate the effects of these agents, even the apparently contradictory result that reducing I_{SI} can hyperpolarize the cell in some cases and depolarize it to a tonic beating state in others. We found that the effect of a particular concentration of neurotransmitter (reflected in the percentage reduction or augmentation of 1 or more conductances) was dependent on the initial bias of the cell. The bias of the cell was manipulated by adjusting the external stimulus current. With I_{STIM} set to zero, either increasing $\bar{g}_R$ by 50% to simulate the effect of low concentrations of 5-HT or reducing $\bar{g}_{SI}$ by 75% to simulate the effect of dopamine or FMRFamide induced a silent hyperpolarized state. These results agree with the reported effects of these neurotransmitters (Benson and Adams 1987). Further, when I_{STIM} was changed to 1.22 nA in the control (see Fig. 9*A*) to simulate a cell with a different initial bias, increasing $\bar{g}_R$ induced a more hyperpolarized bursting state (Fig. 9*C*), which agrees qualitatitively with Fig. 6.8 of Benson and Adams (1987) on the effect of low concentrations of 5-HT. Moreover, in this case (I_{STIM} = 1.22 nA) the reduction of I_{SI} by 50% caused the conversion of a bursting cell to a slow beater (Fig. 9*B*), an effect similar to that observed by Levitan and Levitan (1988a) in response to an analogue of cGMP. Such an effect could also be produced by setting $\bar{g}_{NS}$ to zero. With I_{STIM} set to 1.22 nA, the effects of ELH or high concentrations of 5-HT were also simulated by increasing $\bar{g}_R$ by 50% and increasing $\bar{g}_{SI}$ by 100%. These values led to a type of bursting activity in which both the depolarizing and hyperpolarizing phases were enhanced (see Fig. 9*D*).

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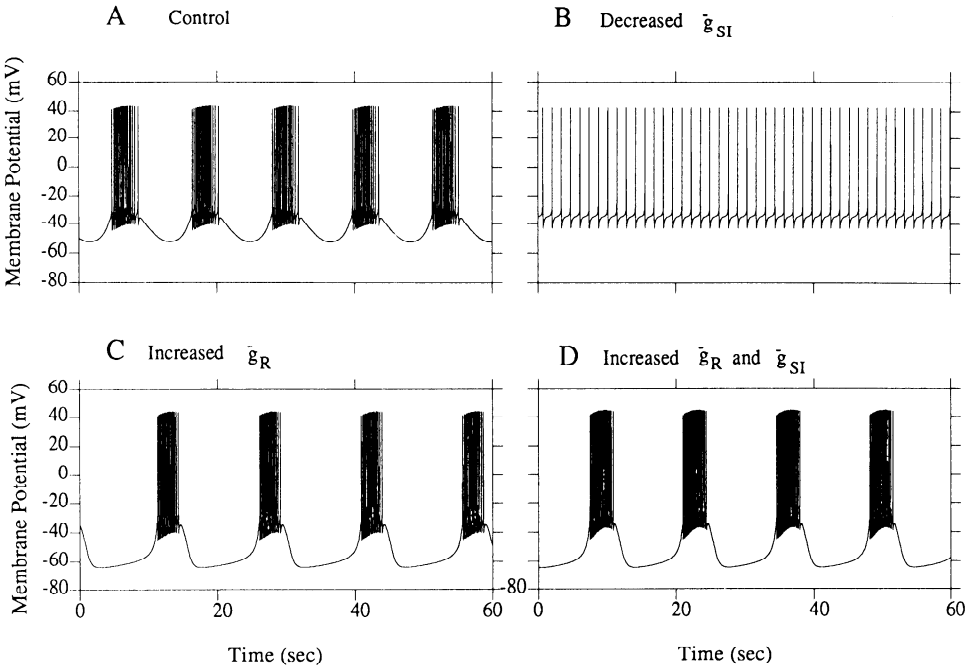


FIG. 9. Effects of modulators. *A*: a control simulation generated with I_{STIM} set to 1.22 nA. *B*: $\bar{g}_{SI}$ was reduced by 50% to simulate the slow beating mode induced by the application of cGMP, which mediates the effects of dopamine. *C*: $\bar{g}_R$ was increased by 50% to simulate the hyperpolarizing effect of the application of low concentrations of 5-HT. *D*: $\bar{g}_R$ was increased by 50%, and $\bar{g}_{SI}$ was increased by 100% to simulate the augmented bursting induced by the application of ELH or high concentrations of 5-HT.

DISCUSSION

The model of R15 simulated bursting, beating, and slow-wave patterns of activity. Moreover, it predicted the transitions between modes of electrical activity as well as the effects of certain modulatory agents.

Modulatory mechanisms

The analyses of both the nullclines and *I-V* relationships described in the RESULTS provide some insight into the mechanisms by which modulatory agents affect R15. In our model the net increase in outward current within the negative slope region of the *I-V* curve caused by an increase in I_R produces an effect on the activity of R15 that is very similar to that of a hyperpolarizing stimulus current. For example, note that the panel labeled I_{STIM} = 1.22 nA in Fig. 5*A* is repeated in Fig. 9*A*, and that the effect on the activity of shifting the stimulus current in a more hyperpolarizing direction (I_{STIM} = 0 in Fig. 5*A*) or of increasing $\bar{g}_R$ (Fig. 9*C*) is nearly the same. Similarly, a decrease in I_{SI} produces an effect that at first glance resembles that of a hyperpolarizing stimulus current. This apparent parallel results from the similar effects on the steady-state *I-V* curve in the vicinity of the applicable fixed points. The parallel breaks down, however, because I_R and I_{SI} are voltage dependent. Thus modulation of these currents does not merely shift the *I-V* curve a fixed amount, like a constant stimulus current, but instead changes its shape. Either a reduction in I_{SI} or an increase in I_R can cause a transition from bursting activity, with the model initially biased at I_{STIM} = 0, to a silent mode (see RESULTS). However, a reduction in I_{SI} tends to damp out the unstable region of negative resistance in the *I-V* relationship, whereas an increase in I_R slightly enhances it. Normal bursting can be restored in the case where $\bar{g}_R$ was increased to simulate the effect of low concentrations of 5-HT (Fig. 9*C*), but not in the case where I_{SI} was decreased to simulate the effect of dopamine (not shown). Thus the model suggests that a cell treated with low concentrations of 5-HT can still be induced to burst if it is biased in the region of negative slope, whereas the *I-V* relationship of a cell treated with dopamine (reduced I_{SI}) is characterized by a reduced region of negative slope and cannot easily be induced to burst.

Another important distinction can also be made between the modulation of I_R and I_{SI} because I_{SI} affects the Ca^{2+} balance, whereas I_R does not. For example, if $\bar{g}_{SI}$ is decreased while all other parameters are unchanged, at any given Ca^{2+} concentration, the Ca^{2+} influx due to I_{SI} will be reduced at all potentials. The calcium nullcline shown in Fig. 6 will shift to the left because equilibrium with respect to Ca^{2+} will be reestablished for each Ca^{2+} concentration at a more depolarized potential where increased voltage-dependent activation of I_{SI} offsets the decrease in conductance. This shift increases the likelihood that the calcium nullcline will intersect the potential nullcline above the spiking threshold in a region that is stable with respect to the slow dynamics of the system and produce beating activity. Thus the effect of changing I_{SI} cannot be predicted simply in terms of its effect on the steady-state *I-V* relationship.

Although the modulation of I_R and I_{SI} is sufficient to account for the experimentally described effects of the above-mentioned modulatory agents, the model simulations show that the modulation of I_{NS} could also account for the effects of GMP. It has been suggested that this current is also modulated by GMP, either directly, or indirectly via decreased influx of Ca^{2+} due to reduced Ca^{2+} current (Levitan and Levitan 1988a).

Comparison with previous models

Previous experimental analyses of the membrane currents that underlie bursting activity in R15 and other cells have focused on two currents (or 2 families of currents) that can be designated as I_D (depolarizing) and I_H (hyperpolarizing). I_D is responsible for boosting spiking frequency during the early part of a burst and for the DAP,

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whereas I_H is responsible for decreasing spiking frequency later in a burst and for burst termination. Because these currents are difficult to observe in isolation (especially at depolarized potentials where large-amplitude transient currents are also active), they have been studied by the use of measurements of the steady-state *I-V* relationship (Gola 1974; Gorman and Hermann 1982) (either with slow voltage ramps or long voltage-clamp steps) or of "tail currents" (Adams and Levitan 1985; Kramer and Zucker 1985; Lewis 1984; Smith and Thompson 1987), after either voltage-clamp pulses or after a freely bursting cell is voltage clamped at various points during a burst. Unfortunately, currents measured in different ways and in different cells have been given various names. It is difficult to determine whether these currents arise from identical conductance mechanisms, although the term slow inward current (SIC) has been associated with currents measured by the use of both methods.

For example, the steady-state *I-V* technique has been used to examine a current (designated I_{NSR}) responsible for the negative slope resistance region in the *I-V* relationship. This current has been associated with the SIC measured during long voltage-clamp pulses in R15 (Gola 1974; Lewis 1984) because of the similarity in their voltage dependence of activation. [See also Adams and Benson (1985) who suggest that I_{NSR} has 2 components, 1 fast and 1 slow. In their view, the slow component is only partially responsible for I_{NSR}.] On the other hand, the SIC in R15 (Gola 1974; Lewis 1984), on the basis of comparable voltage dependence of activation, has also been tentatively identified with a current (termed I_B) measured by the use of the tail current method on bursting cells in *Tritonia* (Adams et al. 1980; Smith and Thompson 1987). I_B has a slow time constant of activation at hyperpolarized potentials and a relatively fast one at more depolarized potentials, enabling the level of activation to summate during a burst. We have included the current termed I_{SI} in our model and given it the attributes of both I_{NSR} and I_B. One potential conductance mechanism for I_D in both R15 and in our model, therefore, is the summated voltage activation of I_{SI}. However, alternative mechanisms have been proposed for the generation of I_D. Adams and Levitan (1985), with the use of the tail-current method, observed a threshold activation of I_D and concluded that it was activated by action potentials generated in poorly clamped axonal regions, perhaps by the passive spread of current. On the other hand, Lewis (1984) observed a graded activation of I_D in R15 similar to that observed for I_{NSR}. Lewis also suggested that I_D was calcium activated because it was blocked by internal injection of the Ca^{2+} chelator ethylene glycol-bis(β-amino-ethyl ether)-*N,N,N',N'*-tetraacetic acid (EGTA). In the left upper quadrant bursting (LUQB) cells, Zucker (1988) attributes I_D to a voltage-independent, calcium-activated nonspecific cation current (I_{NS}). Recently, I_D in R15 has also been attributed to calcium-activated nonselective cation channels located on the axon (Levitan and Levitan 1988a). Thus we included I_{NS}, as well as I_{SI}, in our model.

Various formalisms of I_H and I_D have been incorporated into previous models of bursting activity. Many early models (Chay 1983; Hindmarsh and Rose 1984; Plant 1981; Plant and Kim 1976) were based on the premise that I_H represents an increase in an outward current (generally identified as the Ca^{2+}-activated K^+ current, $I_{K,Ca}$). This hypothesis was based on the work of Gorman and Thomas (1980), who found that $I_{K,Ca}$ was linearly dependent on incremental intracellular Ca^{2+} concentration and that it increased *e*-fold for each 24 mV. Thus $I_{K,Ca}$ would be expected to be gradually activated during a burst. Although Adams and Levitan (1985) were able to elicit a long-lasting K^+ current by injecting amounts of Ca^{2+} similar to previous studies (Gorman and Thomas 1980), they were unable to detect a long-lasting component of $I_{K,Ca}$ in a normally bursting R15 or in response to a train of depolarizing pulses. In particular, the long-lasting outward current, which is incremented by each action potential until it is large enough to terminate a burst then gradually decays until a new burst can be initiated, was designated I_H (for hyperpolarizing) (Adams 1985; Adams and Levitan 1985). These studies showed that I_H *1*) is unaffected by concentrations of tetraethylammonium (TEA) that block $I_{K,Ca}$, *2*) does not reverse at the K^+ equilibrium potential, *3*) is unaffected by changes in external K^+ concentration but is dependent on external Ca^{2+} concentration, and *4*) can be blocked by Ca^{2+} channel blockers (Adams et al. 1980). Therefore Adams and Levitan (1985) concluded that the increase in the outward I_H in R15 actually results from a Ca^{2+}-dependent inactivation of the steady-state voltage-dependent Ca^{2+} current. [We have modeled this current as I_{SI}, but see Adams and Benson (1985), who suggest that it is a fast component of I_{NSR} that is inactivated.] Kramer and Zucker (1985) concluded that inactivation of the steady-state Ca^{2+} current was also responsible for I_H in the bursting L_2–L_6 cells in *Aplysia*. Adams and Levitan (1985) further concluded that insufficient Ca^{2+} enters the cell during normal electrical activity to activate a persistent $I_{K,Ca}$. It has also been suggested that the K^+ conductance activated by Ca^{2+} injection is not identical to that activated by Ca^{2+} entry through the membrane (Johnson 1986). The data of Adams and Levitan (1985) do not rule out a role for $I_{K,Ca}$ in the repolarization of the action potential. Such a role seems plausible in view of the reported similar voltage dependence and kinetics of I_K and $I_{K,Ca}$ in R15 (Adams and Benson 1985); the two currents are distinguished only by the Ca^{2+} activation of the latter. Because recent experimental data indicate that $I_{K,Ca}$ is not critical to the bursting activity in R15, we chose not to model this current. Indeed, our model generates realistic simulations of bursting activity without $I_{K,Ca}$.

Several recent models are based on the premise that I_H is due to the inactivation of a calcium current and focus on how the interplay between I_H and I_D can generate typical bursting patterns. The model of Adams and Benson (1985) duplicated many characteristics of a typical burst. Although the descriptions of the fast membrane currents were based on molluscan data, the slow currents (I_D and I_H) were modeled nonphysiologically. I_D was attributed to electrotonic spread of current, and I_H to Ca^{2+}-dependent inactivation of a persistent calcium current. Because the model equations assumed space-clamped conditions and Ca^{2+} was not included as a variable, the slow currents were modeled simply as current generators that were incremented a fixed amount by each spike and decayed exponentially. Minimal, qualitative models by Rinzel and Lee (1987), Chay and Cook (1988), and Chay (1990) describe I_D and I_H as the slow

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activation and even slower inactivation, respectively, of a single current (I_{SI}) and produce parabolic bursting as first activation, then inactivation dominates. The models of Rinzel and Lee and of Chay and Cook include Ca^{2+} as a system state variable and model I_H as due to Ca^{2+}-dependent inactivation. On the other hand, Chay (1990) modeled I_H as due to voltage-dependent inactivation of the slow inward current. In our model, I_D is represented by activation of I_{SI} and I_{NS}, and I_H is represented by the inactivation of I_{SI}. The model presented in this paper is based on experimental data to a greater extent than the above-mentioned models and includes a material balance on Ca^{2+}.

Recently, multiple channel models of excitable cells that incorporate membrane ionic pumps and exchangers as well as a material balance on several ionic species (e.g., Ca^{2+}, K^+, Na^+) have been published that significantly extend the explanatory range of cellular models. This has been done to the greatest extent in modeling of cardiac cells (e.g., DiFrancesco and Noble 1985; Noble et al. 1989; Rasmusson et al. 1990a,b), but in some neuronal models as well (e.g., Yamada et al. 1989). The additional complexity of our model as compared with previous models of R15 is justified in that it serves to provide better, physically based descriptions of current characteristics and function.

Radial variation in calcium concentration

The dynamics of intracellular Ca^{2+} concentration are generally considered important to the kinetics of Ca^{2+}-regulated membrane processes. To adequately characterize the spatial and temporal changes in $[Ca]_i$, methods of measurement must be able to provide spatial resolution of $[Ca]_i$ on the order of microns and time resolution on the order of milliseconds. Although some early measurements were made on diffusion of Ca^{2+} within R15 and other molluscan neurons (Gorman et al. 1984; Nasi and Tillotson 1985) and some neuronal models have included descriptions of 3-D concentric shells (Sala and Hernandez-Cruz 1990; Yamada et al. 1989), sufficiently accurate data on transients of intracellular Ca^{2+} are not yet available to develop an appropriate 3-D model of R15.

In the absence of quantitative data on Ca^{2+} transients of adequate spatial and temporal resolution, we have chosen to model the intracellular Ca^{2+}-accessible volume as a single well-stirred compartment, without detailed treatment of either diffusional delays or spatial nonuniformities. Although this simple lumped parameter model represents an approximation to the actual situation, it greatly reduces the computational complexity of our model without apparently detracting from its illustrative and predictive capability. This approximation is further justified by the possibility that the concentration gradients in R15 are short lived, particularly with respect to burst modulation with its time scale on the order of tens of seconds. Therefore it is not evident that the addition of intracellular shell compartments at this time would be justified in terms of improved predictive ability of the model.

Model limitations

Although our model reflects the current state of knowledge of R15, several aspects of the membrane currents as well as pumps, exchangers, and Ca^{2+} regulation require further investigation. Indeed, one outcome of this modeling study was to highlight those areas for which additional data are required. For example, more experimental data are necessary on the properties of $I_{K,Ca}$. In addition, there is a lack of voltage-clamp information at hyperpolarized potentials for the fast subsystem, including I_{Na}, I_{Ca}, and I_K. These data are important because the ability of the model to produce sustained spiking activity depends critically on the value of the activation time constants at hyperpolarized potentials. Moreover, additional data are required on the currents that affect the Ca^{2+} balance: I_{SI}, I_{NaCa}, and I_{CaP}. Although these currents are orders of magnitude smaller than the spiking currents, they can determine the mode of activity of the model by virtue of their contribution to the location of the calcium nullcline and thus the fixed point of the system.

Our lumped parameter model assumes that the membrane channels are distributed uniformly throughout the soma. However, the nonspecific cation channels are believed to be located on the axon (Levitan and Levitan 1988a), and a role in burst modulation has been proposed for the retrograde electrotonic spread of current produced by action potentials propagating away from the cell body along the axon (Adams 1985; Adams and Levitan 1985). If this is indeed the case, additional information would be required to develop a distributed model that incorporates spatial variation in the density and types of membrane ion channels.

Summary

Despite the limitations inherent in a lumped parameter formulation, this model predicts many features of the activity of R15. In addition, our analysis of the slow dynamics of the model provides insights into the ability of R15 to exhibit different modes of activity. The inclusion of a Ca^{2+} balance is crucial to this aspect of the model. The model also suggests which parameters are critical for determining the characteristics of bursting activity. Previous analyses of the effects of neuromodulators on R15 (Benson and Adams 1987) focused on the potential nullcline, although the descriptions were phrased in terms of the steady-state *I-V* relationships. On the basis of our model simulations, we propose that there are other potential mechanisms by which modulatory agents can exert their effects on electrical activity, including *1*) a change in the slow dynamics of the cell caused by either a shift in the Ca^{2+} nullcline (by acting on a current crucial to the Ca^{2+} balance) or a shift in the potential nullcline (or both in concert) and *2*) by acting on a current (e.g., I_{NS}) that is only significantly activated by action potentials.

APPENDIX

This appendix contains the equations and values of parameters for the model. In these equations, time is in milliseconds, current is in nanoamperes, potential is in millivolts, and concentrations are in millimoles.

Inward currents

I_{Na}: FAST SODIUM CURRENT

$$I_{Na} = \bar{g}_{Na} m^3 h (V - E_{Na})$$

$$m_\infty = \frac{1}{1 + \exp[(-17.23 - V)/5.53]}$$

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$$\tau_m = 1/(A_m + B_m)$$

$$A_m = \frac{0.08(V + 13.0)}{1 - \exp[(-V - 13.0)/4.09]}$$

$$A_h = 0.112 \exp[(-30.0 - V)/10.0]$$

$$\dot{m} = (m_\infty - m)/\tau_m$$

$$\dot{h} = (h_\infty - h)/\tau_h$$

$$h_\infty = \frac{1}{1 + \exp[(V + 22.0)/3.0]}$$

$$\tau_h = 1/(A_h + B_h)$$

$$B_m = 2.15 \exp[(-35.0 - V)/4.01]$$

$$B_h = \frac{0.23}{\exp[(9.65 - V)/23.9] + 1}$$

I_{Ca}: FAST CALCIUM CURRENT

$$I_{Ca} = \bar{g}_{Ca} d^2 f (V - E_{Ca})$$

$$d_\infty = \frac{1}{1 + \exp[(10.0 - V)/3.8]}$$

$$\tau_d = 1/(A_d + B_d)$$

$$A_d = \frac{0.0063(V + 10.81)}{1 - \exp[(-V - 10.81)/5.03]}$$

$$A_f = 0.00325 \exp[(10.0 - V)/7.57]$$

$$\dot{d} = (d_\infty - d)/\tau_d$$

$$\dot{f} = (f_\infty - f)/\tau_f$$

$$f_\infty = \frac{1}{1 + \exp[(V + 20.0)/4.0]}$$

$$\tau_f = 1/(A_f + B_f)$$

$$B_d = 0.04 \exp[(14.08 - V)/5.09]$$

$$B_f = \frac{0.029}{\exp[(20.29 - V)/5.4] + 1}$$

I_{SI}: SLOW INWARD CALCIUM CURRENT

$$I_{SI} = \bar{g}_{SI} s (V - E_{Ca}) \frac{K_{M,SI}}{K_{M,SI} + [Ca]_i}$$

$$\tau_s = 1/(A_s + B_s)$$

$$A_s = \frac{0.0014(V - 54.0)}{1 - \exp[(-V + 54.0)/12.63]}$$

$$\dot{s} = (s_\infty - s)/\tau_s$$

$$s_\infty = \frac{1}{1 + \exp[(-40.0 - V)/10.0]}$$

$$B_s = 0.00013 \exp[(-11.32 - V)/16.8]$$

I_{NS}: NON-SPECIFIC CATION CURRENT

$$I_{NS} = \bar{g}_{NS} b (V - E_{NS}) \frac{[Ca]_i}{[Ca]_i + K_{M,NS}}$$

$$b_\infty = \frac{1}{1 + \exp[(-15.0 - V)/3.0]}$$

$$\dot{b} = (b_\infty - b)/\tau_b$$

$$\tau_b = 500\left(\frac{0.80}{1 + \exp[(10.0 + V)/3.0]} + 0.20\right)$$

I_L: LEAKAGE CURRENT

$$I_L = \bar{g}_L (V - E_L)$$

Outward currents

I_K: DELAYED RECTIFIER

$$I_K = \bar{g}_K n^4 l (V - E_K)$$

$$n_\infty = \frac{1}{1 + \exp[(3.65 - V)/14.46]}$$

$$A_n = \frac{0.0035(V + 17.0)}{1 - \exp[(-V - 17.0)/3.0]}$$

$$l_\infty = \frac{1}{1 + \exp[(32.5 + V)/12.7]}$$

$$\dot{n} = (n_\infty - n)/\tau_n$$

$$\dot{l} = (l_\infty - l)/\tau_l$$

$$\tau_n = 1/(A_n + B_n)$$

$$B_n = 0.04 \exp\left(\frac{-28.0 - V}{10.0}\right)$$

$$\tau_l = 2{,}000\left(\frac{0.90}{1 + \exp[(28.0 + V)/3.0]} + 0.10\right)$$

I_R: ANOMALOUS RECTIFIER

$$I_R = \bar{g}_R \frac{(V - E_K + 5.66)}{1 + \exp\left(\frac{(V - E_K - 15.3)ZF}{RT}\right)}$$

Pumps and exchangers

$$I_{CaP} = \bar{I}_{CaP}\left(\frac{[Ca]_i}{[Ca]_i + K_{M,CaP}}\right)$$

$$I_{NaCa} = K_{NaCa}(DF_{in} - DF_{out})/S$$

$$S = 1 + D_{NaCa}([Ca]_i[Na]_0^r + [Ca]_0[Na]_i^r)$$

$$DF_{in} = [Na]_i^r[Ca]_0 \exp\left(\frac{(r - 2)\gamma VolF}{RT}\right)$$

$$DF_{out} = [Na]_0^r[Ca]_i \exp\left(\frac{(r - 2)(\gamma - 1)VolF}{RT}\right)$$

Internal calcium concentration

$$[\dot{Ca}]_i = \frac{I_{NaCa} - I_{SI} - I_{Ca} - I_{CaP} - \frac{0.197}{\bar{g}_{NS}}\left(I_{NS}\frac{V - E_{Ca}}{V - E_{NS}}\right)}{2Vol_iF} - n[B]_i\dot{O}_C$$

$$\dot{O}_C = k_U[Ca]_i(1 - O_C) - k_R O_C$$

Parameters

$$\bar{g}_{Na} = 30\ \mu S$$

$$\bar{g}_{Ca} = 10\ \mu S$$

$$\bar{g}_K = 70\ \mu S$$

$$\bar{g}_{NS} = 0.5\ \mu S$$

$$\bar{g}_{SI} = 0.747\ \mu S$$

$$\bar{g}_R = 0.32\ \mu S$$

$$\bar{g}_L = 0.1\ \mu S$$

$$R = 8{,}314\ J/kg\ mol\ °K$$

$$T = 295°K$$

$$[Na]_0 = 500\ mM$$

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$[Na]_i = 50$ mM

$[K]_0 = 10$ mM

$[Ca]_0 = 10$ mM

$K_{M,SI} = 25 \times 10^{-6}$ mM

$K_{M,CaP} = 1 \times 10^{-3}$ mM

$K_{M,NS} = 385 \times 10^{-6}$ mM

$\bar{I}_{CaP} = 15.0$ nA

$E_{Na} = 54$ mV

$E_{Ca} = 65$ mV

$E_K = -75$ mV

$E_{NS} = -22$ mV

$E_L = 0$ mV

$C_M = 17.5$ nF

$Vol_i = 4.0$ nl

$F = 96{,}500$ C/mol

$k_U = 100$ mM^{-1} ms^{-1}

$k_R = 0.238$ ms^{-1}

$[B]_i = 112.5 \times 10^{-3}$ mM

$K_{NaCa} = 0.01$

$D_{NaCa} = 0.01$

$\gamma = 0.5$

$r = 4$

$n = 4$

$I_{NaK} = 3.16$ nA

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